Ethernet LAN switching (PART 1)

Physical Layer

18 February 2026 02:37

OSI Model – Physical Layer

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

- Defines physical characteristics of the medium used to transfer data between devices.
- For example, voltage levels, maximum transmission distances, physical connectors, cable specifications, etc.
- Digital bits are converted into electrical (for wired connections) or radio (for wireless connections) signals.
- All of the information in Day 2's video (cables, pin layouts, etc.) is related to the Physical Layer.

Local Area Network - Data Link Layer

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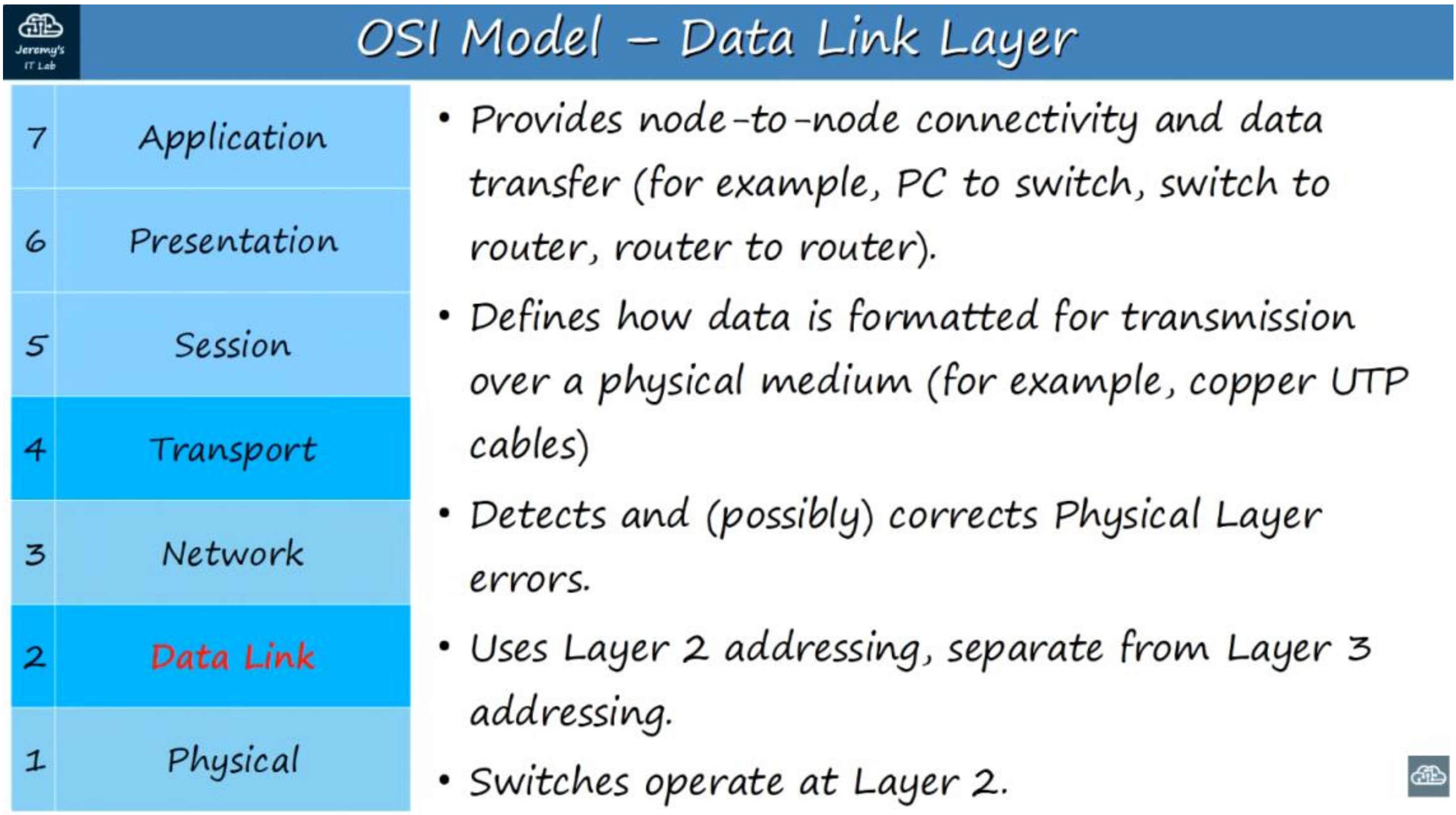

LANs

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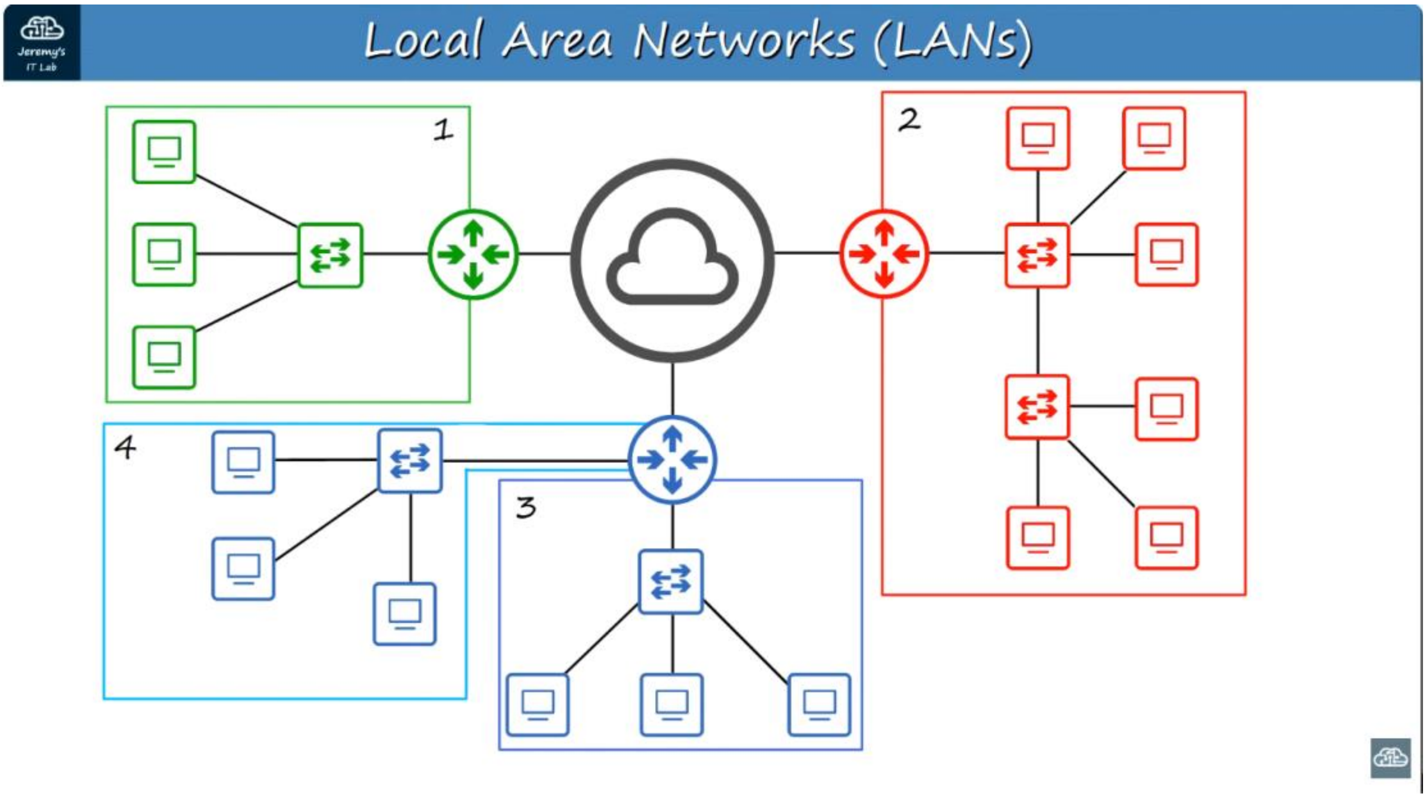

A router has **multiple interfaces** (ports).
Each interface connects to a **different network**.

- Interface 1
- Interface 2
- Interface 3
- Interface 4

Each interface:

- Has its **own IP address**.
- Belongs to a **different subnet/network**.
- Acts as the **gateway** for that LAN.

PDU - Protocol Data Units

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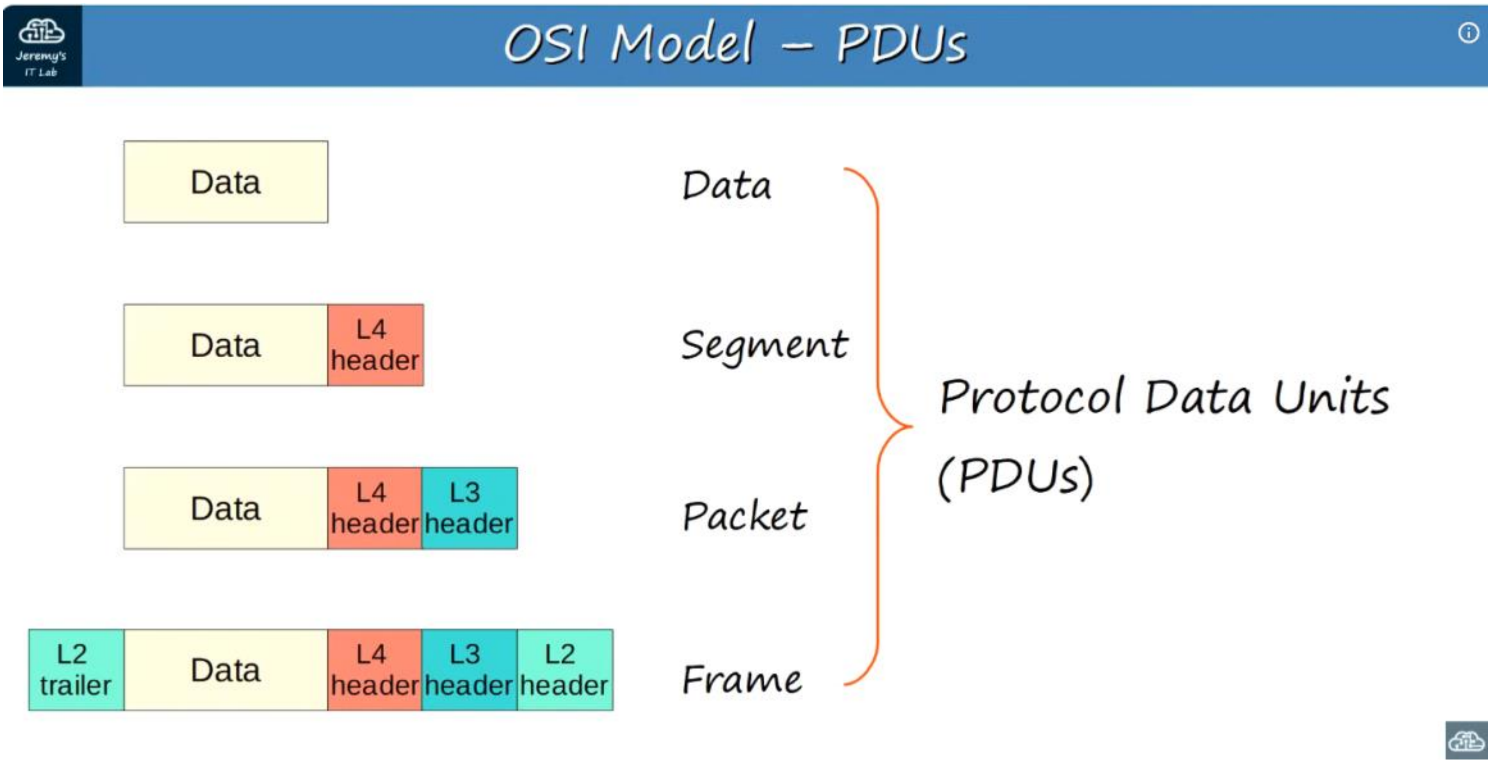

Ethernet Frame

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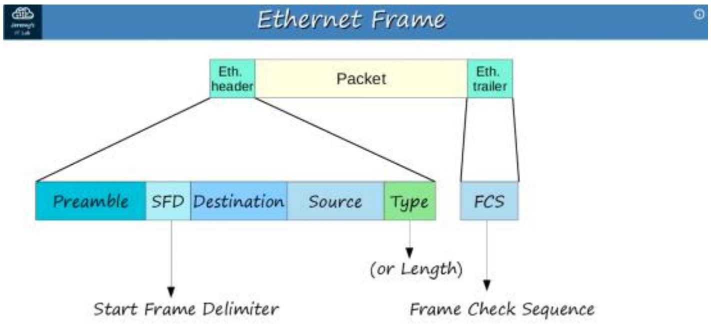

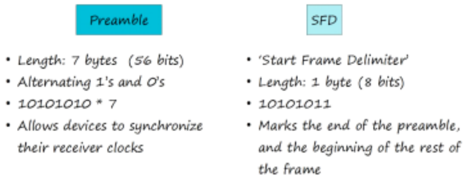

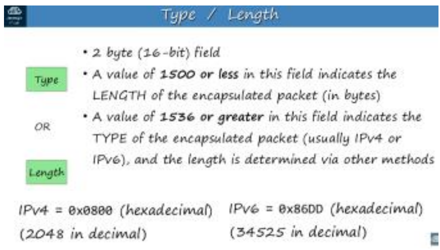

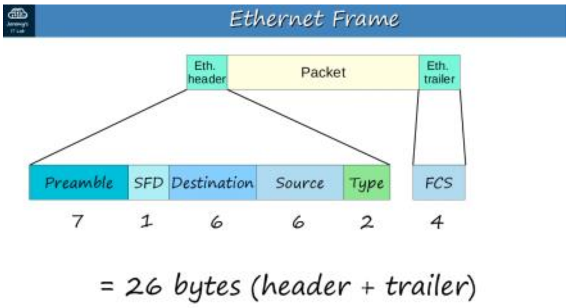

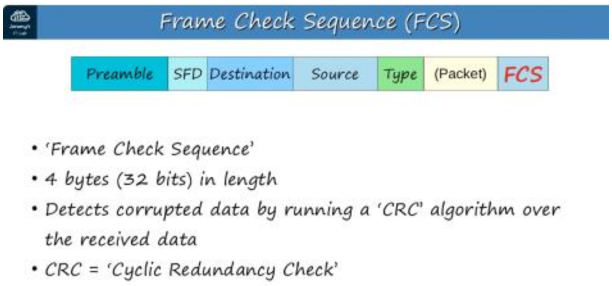

MAC Address

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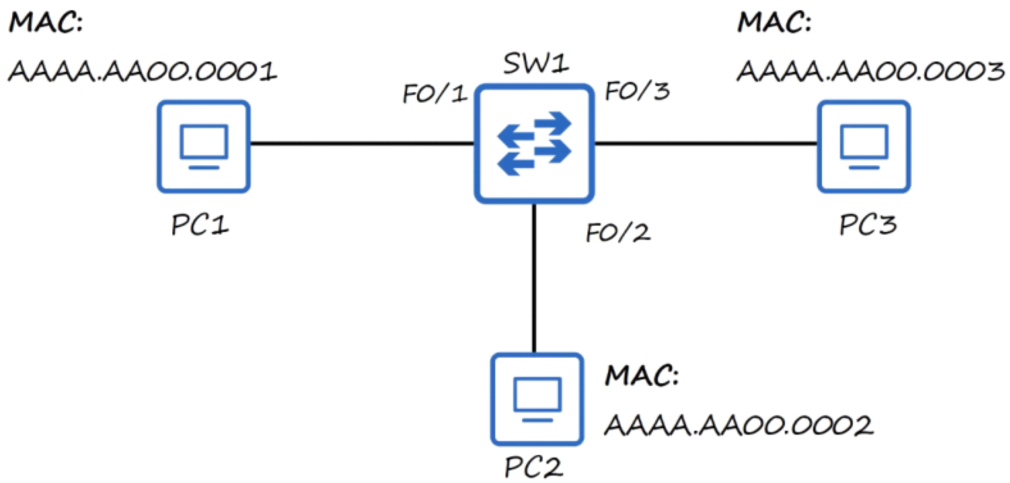
MAC Address

- 6-byte (48-bit) physical address assigned to the device when it is made
- A.K.A. 'Burned-In Address' (BIA)
- Is globally unique
- The first 3 bytes are the OUI (Organizationally Unique Identifier), which is assigned to the company making the device
- The last 3 bytes are unique to the device itself
- Written as 12 **hexadecimal** characters

Hexadecimal

DEC.	HEX.	DEC.	HEX.	DEC.	HEX.	DEC.	HEX.
0	0	8	8	16	10	24	18
1	1	9	9	17	11	25	19
2	2	10	A	18	12	26	1A
3	3	11	B	19	13	27	1B
4	4	12	C	20	14	28	1C
5	5	13	D	21	15	29	1D
6	6	14	E	22	16	30	1E
7	7	15	F	23	17	31	1F

00-0a-83-b1-c0-8e



Types of frame

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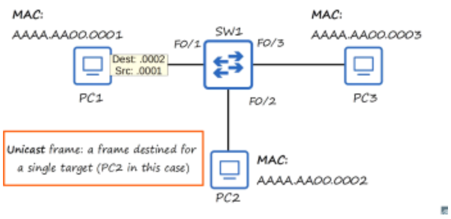
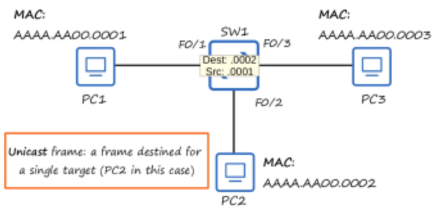
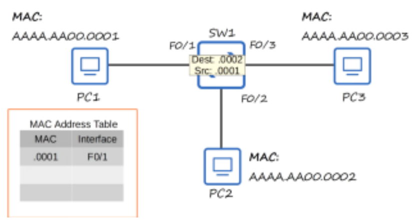
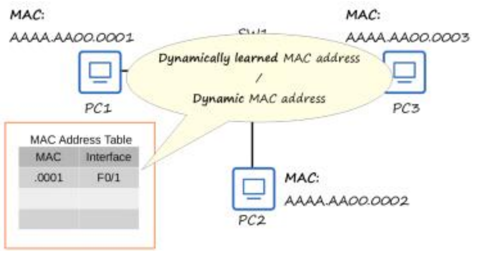
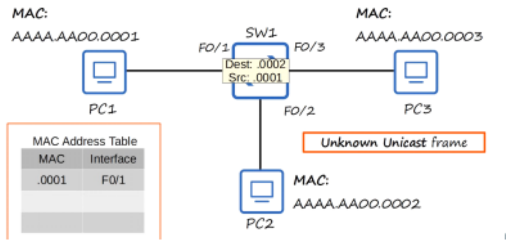
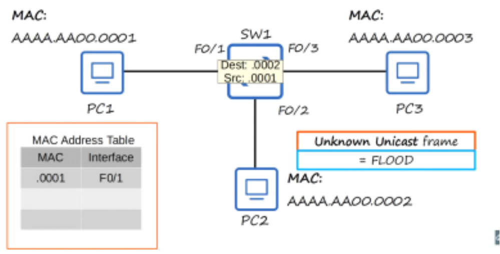


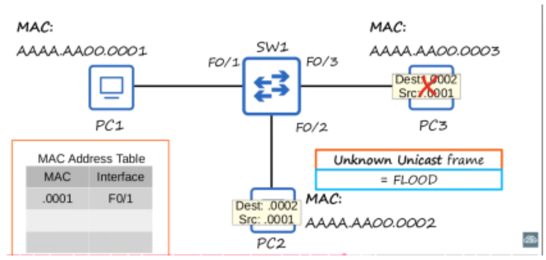


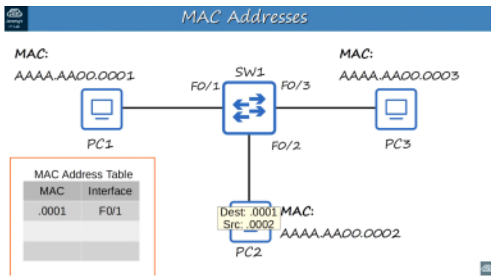


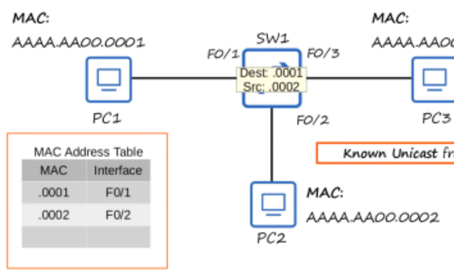


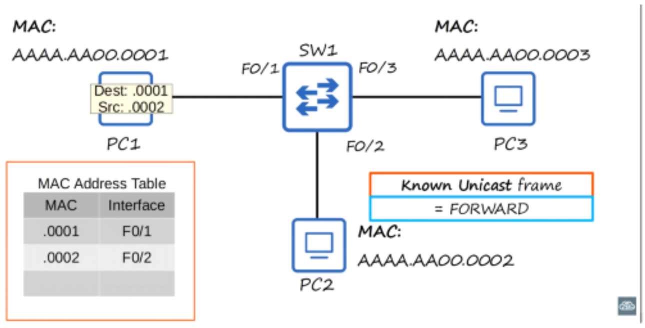


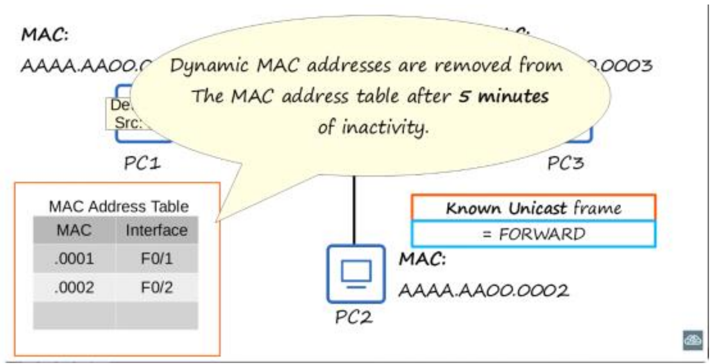












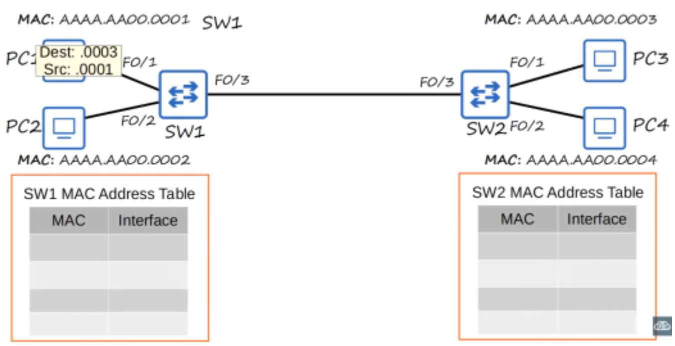
An **unknown unicast frame** is an Ethernet frame where:

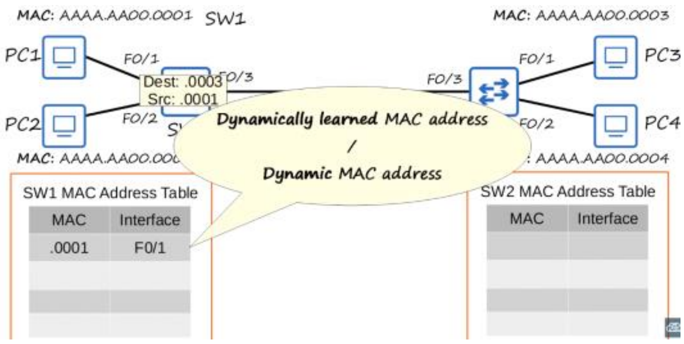
- The **destination MAC address is a specific device** (not broadcast/multicast)
- But the **switch does NOT know which port that MAC address belongs to**

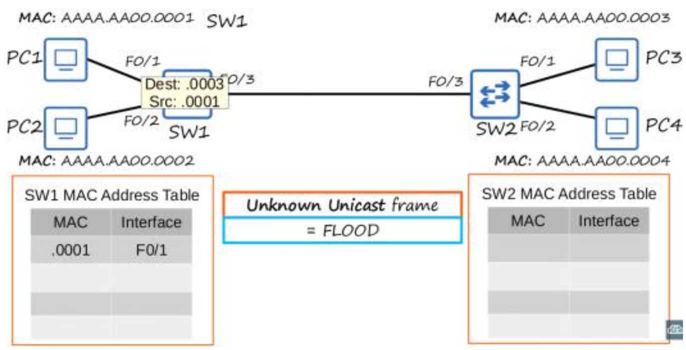
A **known unicast frame** is an Ethernet frame that is sent from one device to **one specific destination device**, and the switch **already knows** which port the destination device is connected to.

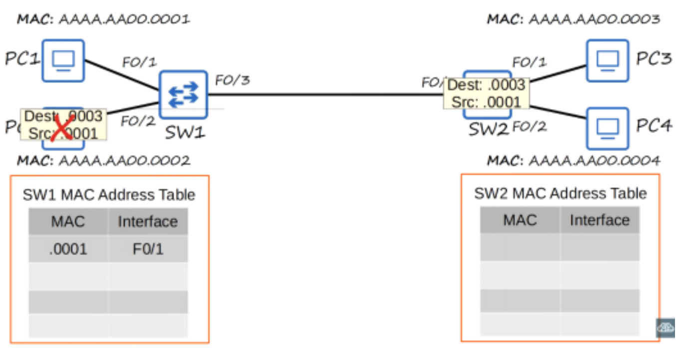
MAC learning & Frame forwarding

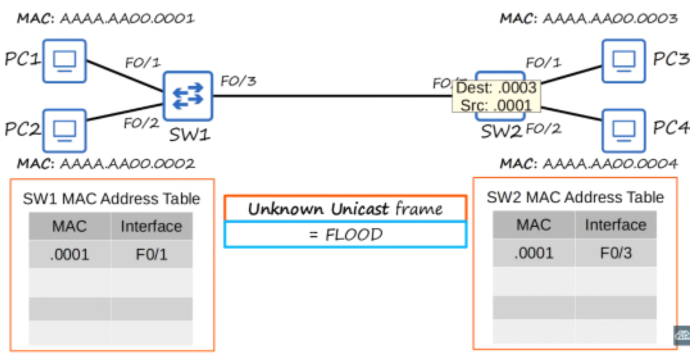
19 February 2026 00:49

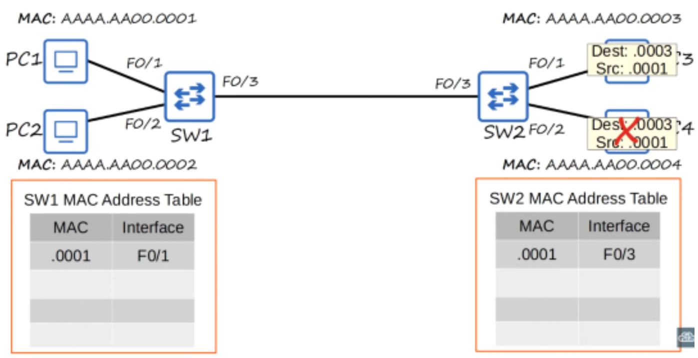


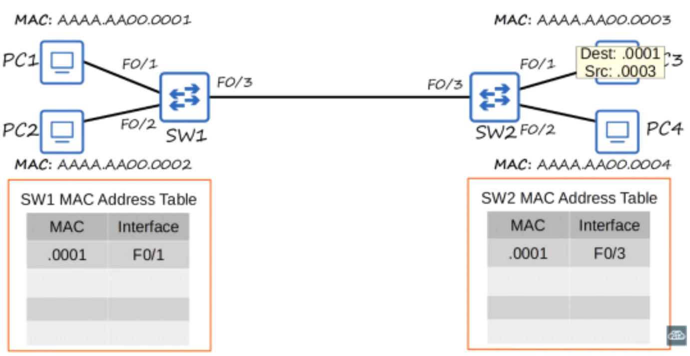


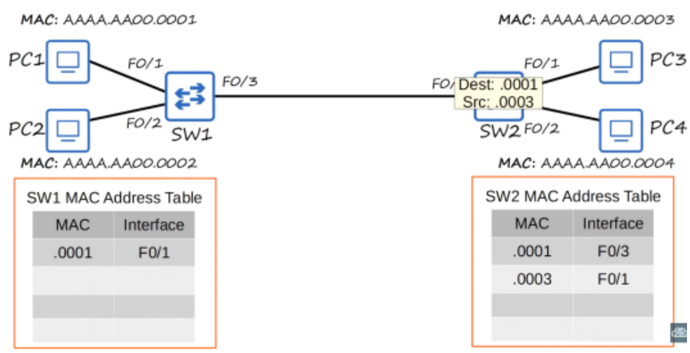


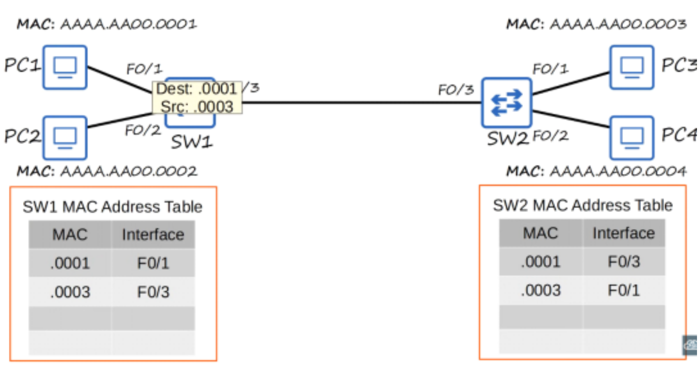


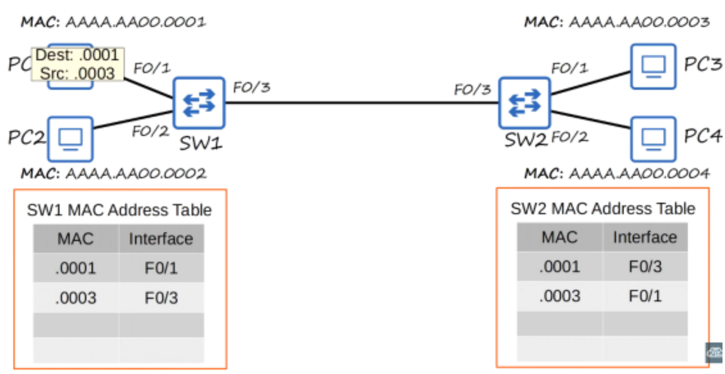












MCQs

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Which field of an Ethernet frame provides receiver clock synchronization?

a) Preamble
b) SFD
c) Type
d) FCS

How long is the physical address of a network device?

a) 32 bytes
b) 32 bits
c) 48 bytes
d) 48 bits

What is the OUI of this MAC address? **E8BA.7011.2874**

a) E8BA
b) E8BA.70
c) 7011
d) E8BA.7011

Which field of an Ethernet frame does a switch use to populate its MAC address table?

a) Preamble
b) Length
c) Source MAC Address
d) Destination MAC Address

What kind of frame does a switch flood out of all interfaces except the one it was received on?

a) Unknown unicast
b) Known unicast
c) Allcast

MCQs Answers

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A) Preamble

D) 48 bits

B) E8BA.70

C) Source MAC Address

A) Unknown unicast